

New Scenario-Based Cumulative Absolute Velocity Models for Shallow Crustal Tectonic Settings

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ABSTRACT

The Pacific Earthquake Engineering Research Center Next Generation Attenuation-West2 database is used to derive a new conditional ground-motion model (CGMM) and a set of scenario-based models for estimating cumulative absolute velocity (CAV) for earthquakes in shallow crustal tectonic settings. Random-effects regressions were performed to develop the conditional model, with random effects across different earthquakes. The estimate of CAV is conditioned on the estimated peak ground acceleration (PGA), the time averaged shear-wave velocity in the top 30 m (V_{s30}), the earthquake magnitude (M_w), and the rupture distance (R_{rup}). By combining the conditional CAV model with ground-motion models (GMM) in shallow crustal earthquake zones for PGA, new scenario-based models are developed for estimating the median CAV and its standard deviation, directly from an earthquake scenario and site conditions. A scenario-based CAV model captures inherently the complex ground-motion scaling effects included in the GMMs for spectral accelerations on which it is based on, such as, sediment-depth effects, soil nonlinearity effects, and regionalization effects. This approach also ensures consistency between the estimated CAV values and a design spectral acceleration response spectrum. The conditional and scenario-based models to estimate CAV are presented, and trends of the developed scenario-based models and previous traditional models for CAV are compared. Interestingly, we found a remarkable consistency between scenario-based and traditional unconditional CAV models, when the underlain spectral GMM used in the implementation of the scenario-based model is properly constrained. Finally, we provide examples for the use of the conditional and scenario-based models in performance-based earthquake engineering.

KEY POINTS

- A new conditional and new scenario-based models for the estimation of CAV are proposed.
- The conditional CAV model can be used to select ground motions, and the scenario-based models are quite robust.
- The implemented scenario-based models for CAV can be readily used in performance-based earthquake engineering.

Supplemental Material

INTRODUCTION

The cumulative absolute velocity (CAV) is a scalar ground-motion intensity measure (IM) that is defined as

$$\text{CAV} = \int_0^{t_{\max}} |a(t)| dt, \quad (1)$$

in which $a(t)$ is the acceleration at time t in units of g , $t_{\max}$ is the duration of the time series (Electric Power Research Institute

[EPRI], 1988). There are other variations of CAV that include minimum thresholds of the acceleration that is included in the integral. For example, EPRI (1988) limited the integral over time to only include 1 s time windows that had a $\text{PGA} \geq 0.025g$. This is called standardized CAV and is denoted as CAV_{std} .

Rather than using 1 s windows, Kramer and Mitchel (2006) defined CAV_5 based on a threshold of $|a(t)| > 5 \text{ cm/s}^2$ for the integral.

When first introduced by EPRI (1988), CAV was used to distinguish between potentially damaging ground motions and nondamaging ground motions. In particular, CAV was used to evaluate the damage potential of ground motions from small and moderate earthquakes with large response spectral

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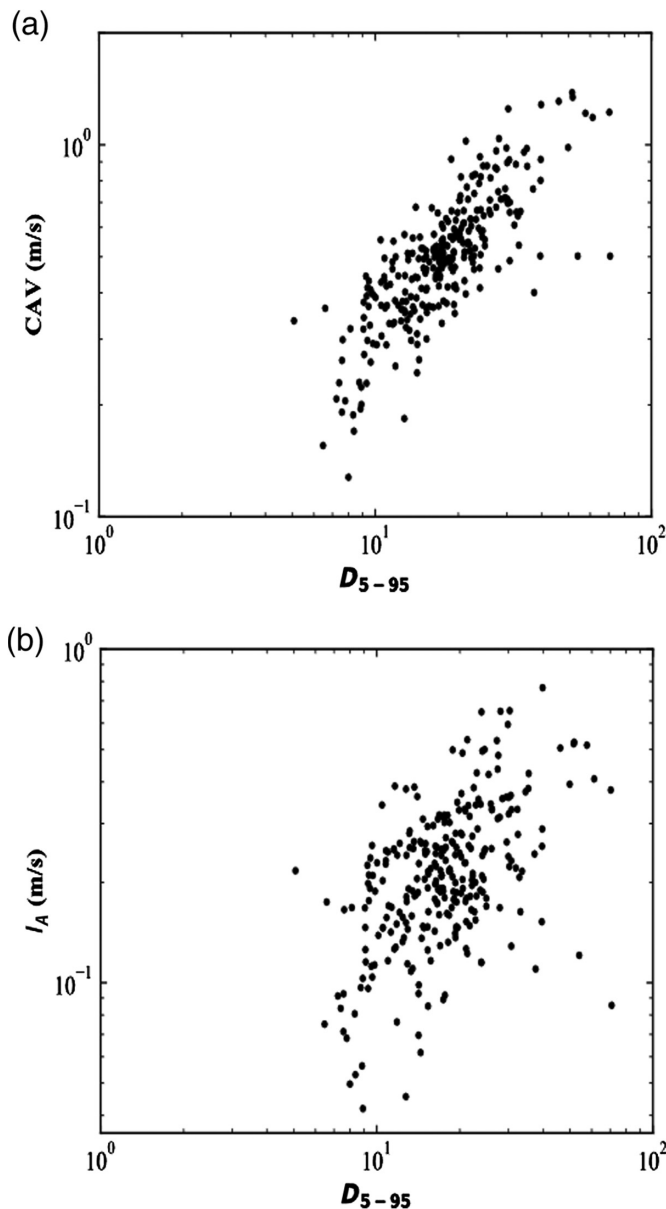


Figure 1. Scaling of (a) cumulative absolute velocity (CAV) and (b) I_A versus D_{5-95} for a subset with M_w from 6.0 to 7.0 and peak ground acceleration (PGA) from 0.10g to 0.15g. Notice that dispersion in the I_A scaling is larger.

values in the high-frequency range but without long enough duration to damage well-engineered structures.

Since its initial use for identifying nondamaging ground motions, CAV has begun to be used as a scalar intensity measure that combines both the amplitude and the duration. For example, in the field of geotechnical engineering, both duration and amplitude of the ground motion are important for soil liquefaction and settlements. [Kramer and Mitchell \(2006\)](#) investigated 300 candidate IMs and found CAV_5 was the most efficient IM to capture the generation of excess pore pressures in soil liquefaction. Similarly, [Kwan et al. \(2018\)](#) found that CAV_5 was the most efficient IM for predicting liquefaction-induced

displacements. CAV has been also used in soil structure interaction problems: for instance, [Macedo \(2017\)](#) and [Bray and Macedo \(2017\)](#) found that both CAV and CAV_{std} are efficient IMs for the estimation of liquefaction-induced building settlements. [Bullock, Dashti, Karimi, et al. \(2019\)](#) and [Bullock, Dashti, Liel, et al. \(2019\)](#) found that CAV, estimated at outcrop conditions, was the most efficient scalar IM for the estimation of liquefaction-induced building settlements and was also effective in estimating tilting.

CAV has also been evaluated for use as an IM for structural analyses and time-history selection. [Muin and Mosalam \(2017\)](#) explored the possibility of using CAV to identify the onset, location, and severity of local damage for structural health monitoring. [Danciu and Tselentis \(2007\)](#) developed fragility functions based on CAV_5 for multiple-story reinforced concrete buildings in Greece. [Katona \(2012\)](#) used CAV in the seismic risk analysis of nuclear facilities. [Fahjan et al. \(2011\)](#) used CAV in the context of earthquake early warning systems. [Tarbali et al. \(2019\)](#) used CAV as one of the IMs for time-history selection. They showed that record selection that considered CAV, 5%–75% duration (D_{5-75}), and 5%–95% duration (D_{5-95}), in addition to spectral accelerations “PSA(T),” led to the appropriate percentage of records with near-field pulses.

Another IM that combines duration and amplitude information is the Arias intensity, I_A , is given by

$$I_A = \frac{\pi}{2g} \int_0^{t_{\max}} a^2(t) dt. \quad (2)$$

Because the I_A is based on the square of the acceleration, it is more sensitive to amplitude and less sensitive to duration than CAV. Figure 1a,b shows the dependence of CAV and I_A on duration for earthquakes with magnitude (M_w) between 6.0 and 7.0 and PGA values between 0.1g and 0.15g. For this set of ground-motion data, CAV has a stronger dependence on duration than I_A . This suggests that for systems whose response depends on number of cycles above some amplitude threshold, CAV is expected to be a more efficient IM than I_A .

CURRENTLY AVAILABLE CAV GMMs

Given the growing applications for CAV, previous research efforts have developed ground-motion models (GMMs) to estimate CAV (e.g., [Campbell and Bozorgnia, 2010, 2012, 2019](#); [Du and Wang, 2012](#); [Foulser-Piggott and Goda, 2015](#)). Most of the available CAV models were developed as traditional GMMs that directly estimate CAV, given the earthquake characteristics (e.g., magnitude, M_w ; earthquake–site distance, R_{rup} ; fault type; and so forth) and site conditions represented by parameters such as V_{S30} (the time-averaged shear-wave velocity over the top 30 m). For example, the GMM developed by [Campbell and Bozorgnia \(2010\)](#), referred as the CB10 model in this study, was developed using the ground-motion database from the Next Generation Attenuation-West1 (NGA-West1)

project (Chiou *et al.*, 2008) and provides CAV estimates for the geometric mean of two horizontal ground-motion components, using the same earthquake and site input parameters as they used in their GMM for PSA. Du and Wang (2012) simplified the CB10 model by reducing the number of parameters used in the original GMM and achieved similar magnitude and distance scaling as those in the CB10 model in the center of the data range. Later, Foulser-Piggott and Goda (2015) developed a CAV GMM using a Japanese ground motion database, and Bullock *et al.* (2017) developed a CAV model for rock conditions. More recently, Campbell and Bozorgnia (2019) updated their CB10 model using the NGA-West2 ground-motion database (Ancheta *et al.*, 2014); this updated model is referred as CB19 in this study.

A different approach was used by EPRI (2006) to develop a model for CAV_{std} for the central and eastern United States (CEUS). The EPRI (2006) approach included the observed PGA as an input parameter for the GMM. As will be discussed in the next section, this is now called the conditional GMM (CGMM) approach. In 2006, the available ground-motion data from the CEUS was very sparse. By conditioning the CAV_{std} on the observed PGA, the ground-motion data from the western United States could be combined with the CEUS data.

Although there are current GMMs for the various CAV metrics, a key issue for using CAV is that the parameterization and extrapolation of the CAV model should be consistent with the selected $PSA(T)$ models. For example, Du and Wang (2012) took a statistical approach to developing their empirical CAV model and found that a simplified parameterization compared to that used by CB10 was adequate to fit the data. The problem with this simplification is that the scenarios that control hazard are often not well represented in the empirical data. Hanging-wall (HW) effects is a good example: there are few recorded ground motions at sites located over the HW, but the HW effects can lead to a doubling of the short-period ground motions compared with the footwall (FW) sites at the same distance (Donahue and Abrahamson, 2014). Using a simplified model for CAV without HW effects can lead to CAV values that are inconsistent with the short-period PSA values from PSA GMMs that include this important effect. For example, consider the CAV for an M_w 7.5 reverse earthquake with a dip of 30 for sites located at a rupture distance of 5 km on both the HW and FW sides of the rupture. The Du and Wang (2012) model gives a median CAV of 18 m/s for both sites, whereas, the median PGA for the Campbell and Bozorgnia (2014) GMM that includes HW effects are 0.43g and 1.21g for the FW and HW sites, respectively. The single CAV value from Du and Wang (2012) is not consistent with the large difference in the PGA values for sites on the FW and HW for this scenario. In contrast, the CB19 model gives median CAV values of 33 and 21 m/s for the HW and FW sites, respectively, which is consistent with the trend in PGA values. This example shows that simplified CAV models that exclude key physical behaviors can lead

to inconsistent results for scenarios that are outside the range constrained by the empirical data.

The point is that GMMs used in seismic hazard analysis need to capture the important physical behaviors so that they can be reliably extrapolated to the scenarios that control the hazard. This involves more than just simple statistical curve fitting based on the fewest parameters needed to fit the central range of the data. There is a need for CAV GMMs that are consistent with the scaling and extrapolation used for the $PSA(T)$ GMMs. The CB19 CAV model is consistent with scaling in the CB14 PSA GMM, but some aspects of the scaling in the CB14 model are significantly different from the other NGA-West2 PSA GMMs. So, the CB19 CAV model may be inconsistent with the other NGA-West2 GMM for some scenarios. The CGMM for CAV developed in this study is combined with the NGA-West2 PSA GMMs to develop a suite of scenario-based CAV models that are consistent with the individual NGA-West2 GMMs.

ESTIMATING SECONDARY PARAMETERS FOR DESIGN GROUND MOTIONS

In most seismic design, the primary design ground motion is the 5% damped elastic response spectrum for the horizontal component. Therefore, we define PGA and 5% damped spectral accelerations as the primary IMs and all other IM as secondary IMs. In addition to the horizontal response spectrum, other secondary IMs can be important for the response of some structural systems. For example, CAV is important for geotechnical systems that are sensitive to the combined effect of the duration and the amplitude of the ground motion.

There are two main approaches for estimating the secondary ground-motion parameters, given a design earthquake scenario and a horizontal-component design spectrum, $PSA(T)$: the traditional GMM approach and the CGMM approach. The traditional approach develops an empirical GMM for the secondary ground-motion parameters based on M_w , R_{rup} , and site conditions in a manner similar to the approach used to develop GMMs for the horizontal $PSA(T)$ values. For estimating the secondary parameter, results from a probabilistic seismic hazard assessment (PSHA) are used and the representative M_w and R_{rup} of the design earthquake scenario are determined from the deaggregation of the hazard for $PSA(T)$ at the periods important for the structure. The M_w and R_{rup} are then used in GMM for the secondary IM. In practice, the target design values of the secondary IMs are often based on the 16th and 84th percentile values for the controlling scenario from the deaggregation. In some applications, the epsilon of the design $PSA(T)$ is also considered for estimating the secondary parameters.

If the correlation of the residuals from the secondary IM and the $PSA(T)$ at the period of interest is available, then the median and variability of the secondary parameters, conditioned on the $PSA(T)$, can be estimated following the same approach as used for conditional spectra (Baker, 2011). Given the design value of

the primary intensity measure, IM_1 , the mean and standard deviation of the secondary parameter, IM_2 , are given by

$$\text{Mean}(IM_2|IM_1) = \text{Mean}(IM_2|M, R) + \varepsilon_{IM_1}\rho\sigma_{IM_2}, \quad (3)$$

$$\sigma_{IM_2|IM_1}^2 = (1 - \rho^2)\sigma_{IM_2}^2, \quad (4)$$

in which ρ is the correlation between the normalized residuals of IM_1 and IM_2 .

An alternative approach is to directly include the design $PSA(T)$ values as predictive parameters in the GMM for the secondary parameters, in addition to the source, path, and site parameters used in traditional GMMs. Abrahamson *et al.* (2016) and Macedo *et al.* (2019) used the term “CGMM” for GMMs that include other $PSA(T)$ values as model input parameters. This is consistent with the term “conditional” used in the conditional mean spectrum (CMS) described by Baker and Cornell (2006). The CMS approach uses the design response spectral value at the conditioning period as an input to compute the expected response spectral values at other spectral periods. The CGMM approach differs from the CMS approach in that the CGMM uses the design $PSA(T)$ value directly as an input parameter rather than using the epsilon of the design $PSA(T)$ and the correlation coefficient between the epsilon values (normalized residuals) for the two spectral periods.

The CGMM approach estimates the median value of a secondary parameter, IM_{sec} , according to

$$IM_{sec} = c_1 + c_2IM_1 + f(M, R, \text{Site}, \dots). \quad (5)$$

The $f(M, R, \text{Site}, \dots)$ term models the difference in the source, path, and site scaling between the scaled primary IM and the secondary IM. It does not represent the base magnitude, distance, and site scaling of IM_{sec} by itself.

Using propagation of errors, we can compute the standard deviation of IM_{sec} from the standard deviations of the primary IM and the standard deviation of the CGMM:

$$\sigma_{IM_2}^2 = \left(\frac{\partial IM_2}{\partial IM_1}\right)^2 \sigma_{IM_1}^2 + \sigma_{IM_2|IM_1}^2. \quad (6)$$

From the form of the CGMM in equation (5), the partial derivative is just the coefficient c_2 . Therefore, this equation becomes

$$\sigma_{IM_2}^2 = c_2^2 \sigma_{IM_1}^2 + \sigma_{IM_2|IM_1}^2. \quad (7)$$

Substituting from equation (4) for $\sigma_{IM_2|IM_1}^2$ leads to

$$\sigma_{IM_2}^2 = c_2^2 \sigma_{IM_1}^2 + (1 - \rho^2)\sigma_{IM_2}^2. \quad (8)$$

Solving for c_2 leads to

$$c_2 = \rho \frac{\sigma_{IM_2}}{\sigma_{IM_1}}. \quad (9)$$

So, in the CGMM approach, the coefficient on the primary IM contains the correlation between the epsilons for the primary and secondary IMs. It may seem easier to just follow the CMS approach, but there are key advantages of using the CGMM approach.

The first advantage of the CGMM approach is related to consistency with the scaling in the GMMs used for the primary IMs. Using the correlation approach to estimate the secondary parameter (equations 3 and 4) implicitly assumes that the GMM for the secondary parameter has a consistent parameterization and consistent physical constraints on the extrapolation as the PSA GMM. This is the case for spectral accelerations at difference periods from a single GMM, as used for the CMS, but it may not be the case for secondary parameters for two reasons: the GMMs for the secondary parameters are often developed by different researchers than those that developed the $PSA(T)$ GMM; and there is less information available from analytical studies to provide physical constraints on the scaling for the secondary parameters. As a result, the GMM for the secondary parameter may be inconsistent with the PSA GMM in terms of how the model extrapolates outside the empirical data range. The lack of the HW effect in the simplified CAV model developed by Du and Wang (2012) discussed earlier is an example of this. Using the CGMM approach, we can take advantage of the significant effort that has been put into developing constraints on the extrapolation in the PSA GMM scaling using analytical modeling for effects not well constrained by the empirical data such as short-distance saturation, HW effects, soil-depth effects, and nonlinear soil effects.

A second advantage of the CGMM approach is that CGMM are more stable with less regional differences than GMMs developed using the traditional approach. Because the $PSA(T)$ value already includes much of the magnitude, distance, and site scaling effects, CGMMs have weaker dependence on these parameters than traditional GMMs. They also have much smaller aleatory variability. Because of the smaller aleatory variability and the simpler functional forms required, the coefficients of conditional models are stable and well constrained by the available empirical data, which makes the regression more robust.

Third, regional differences in the scaling for the secondary parameters can be easily estimated by combining a global conditional model with region-specific GMMs for the PSA values. Because the conditional models use the PSA values as predictive parameters, they show fewer regional differences than traditional GMMs. That is, the regional differences are captured in the PSA values, so a global ergodic CGMM is more likely to be applicable to many different regions than a traditional GMM.

A single global CGMM can be combined with a suite of GMMs for the median and standard deviation of the PSA values to develop a suite of GMMs for the secondary parameter that fully captures the appropriate aleatory variability as well as the magnitude, distance, and site scaling of the median of the

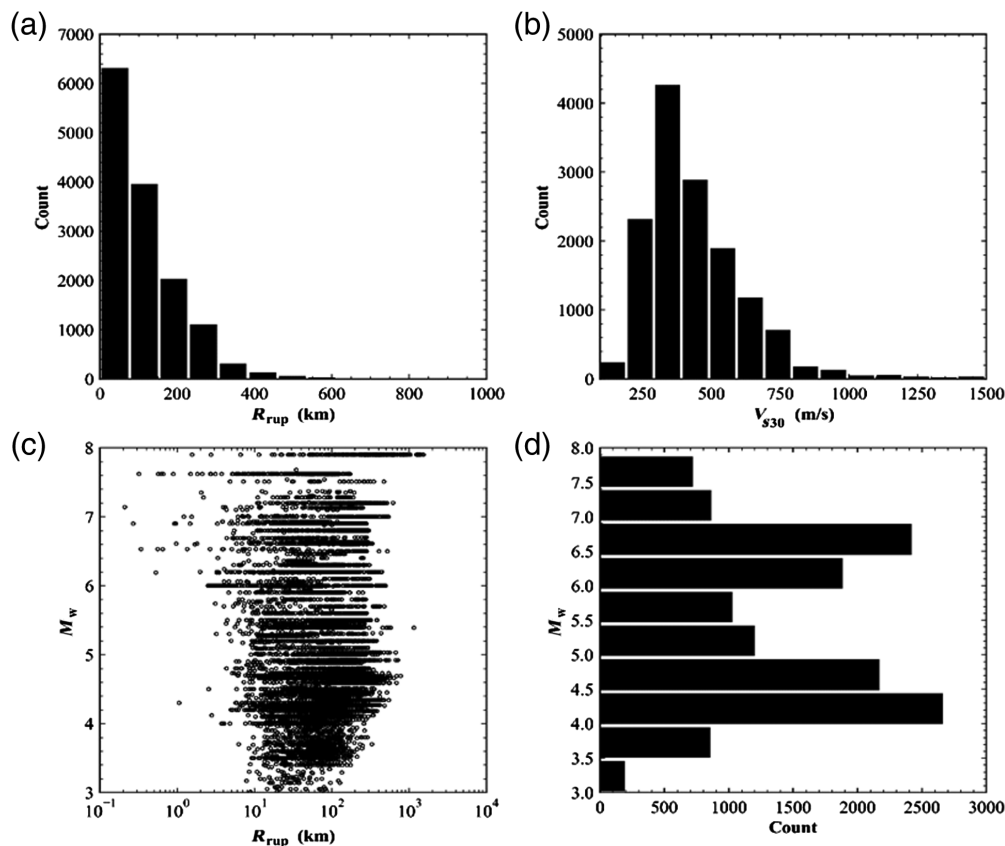


Figure 2. NGA-West2 selected ground motions recordings for this study (a) R_{rup} histogram, (b) V_{s30} histogram, (c) magnitude-distance distribution, (d) M_w histogram.

secondary parameter that is consistent with selected $PSA(T)$ GMMs. This combination of a CGMM and a suite of $PSA(T)$ models has been referred as a scenario-based model by Macedo *et al.* (2019), because the secondary IM and its variability can be estimated for a given earthquake scenario (similar to a traditional GMM). For example, the single conditional model for I_A developed by Macedo *et al.* (2019) was combined with GMMs for the PGA and $PSA(T = 1)$ values in subduction zones to produce five alternative GMMs for I_A . In this approach, the magnitude, distance, and site scaling for each $PSA(T)$ model, including the constraints on the extrapolation, nonlinear site effects and HW effects are incorporated into the new I_A models.

DATA SET

This study used the NGA-West2 database developed by Pacific Earthquake Engineering Research Center (PEER; Ancheta *et al.*, 2014), which contains worldwide shallow crustal earthquakes that occurred until 2011. The full database has 21,540 recordings from 599 earthquakes. We used the following criteria to select the subset for regression: (1) recordings with missing metadata were removed; (2) data with Z_{TOR} (depth to the top of the rupture) larger than 30 km was removed;

and (3) recordings with only one horizontal component were removed.

The final selected subset has 14,000 ground-motion recordings (each recording has two horizontal components) from 287 earthquakes, with magnitudes from 3.0 to 7.9. The magnitude and distance distribution of the selected subset is presented in Figure 2.

BASIS FOR THE FUNCTIONAL FORM IN THE CAV CGMM

In considering the physical basis for a CAV GMM, we first note that the term “CAV” is a misnomer, because the integral in equation (1) defines the cumulative absolute acceleration; however, because it includes the integration step size, it has units of velocity, and, the original developers of this IM thought that it would be confusing, if they defined a parameter called “cumulative absolute acceleration” that had

units of velocity. Although the goal was to avoid confusion, the use of the term CAV leads to misunderstandings, as it sounds like it is related to the frequency content relevant for velocity, but, as discussed later, CAV has little to do with velocity.

The term CAV is well established in the ground-motion field, so we follow the common convention and call this parameter CAV, but when considering how we expect CAV to depend on seismological parameters and ground-motion characteristics, it is important to remember that CAV is the average absolute value of the time history multiplied by the duration and is not sensitive to the low-frequency content. For example, if $a_1(t)$ is a 10 Hz sine wave with unit amplitude, $a_2(t)$ is a 1 Hz sine wave with unit amplitude, and they both have the same duration, then the CAV is the same for $a_1(t)$ and $a_2(t)$, even though their velocities will differ by a factor of 10. As another example, Figure 3 shows the Fourier spectra of acceleration for two different cases: case 1 has corner frequency of 1 Hz, and case 2 has a corner frequency of 0.2 Hz. Although, there is a large increase in the low-frequency content for case 2, the CAV only increases by 5%. This shows that CAV is a high-frequency parameter that is only weakly affected by the low-frequency content controlling the velocity range.

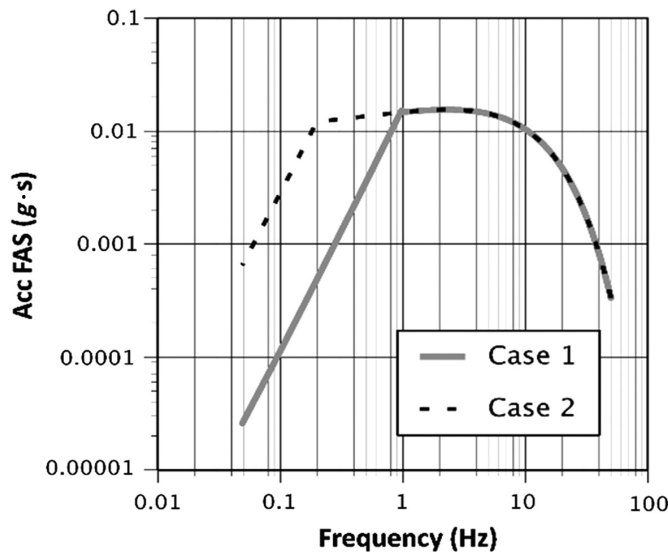


Figure 3. Acceleration Fourier spectra (AccFAS) for two different cases: case 1 has a corner frequency of 1 Hz, and case 2 has a corner frequency of 0.2 Hz. Although, there is a large increase in the low-frequency content for case 2, the CAV only increases by 5%.

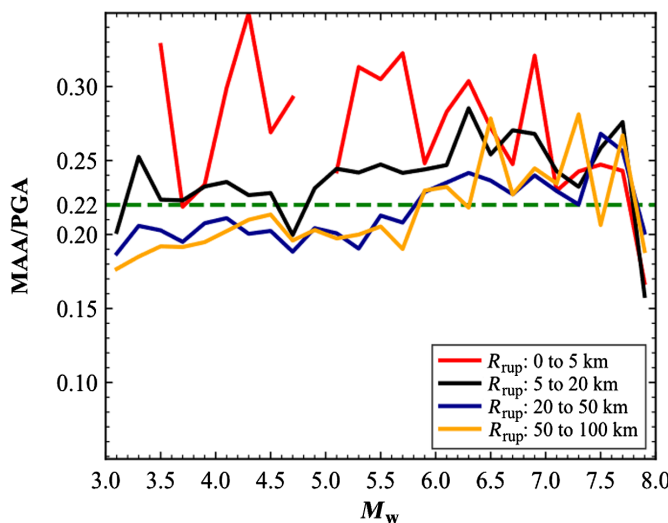


Figure 4. Mean absolute acceleration (MAA)/PGA ratio as a function of magnitude for different distances. The horizontal line at MAA/PGA = 0.22 corresponds to the point-source stochastic model. The color version of this figure is available only in the electronic edition.

CAV can be written as the product of the mean absolute acceleration (MAA) and the duration (see equation 1):

$$\ln(\text{CAV}) = \ln(\text{MAA}) + \ln(t_{\max}), \quad (10)$$

in which MAA is the mean absolute acceleration, and $t_{\max}$ is the duration. As a high-frequency parameter, the MAA will be strongly correlated with the PGA. Campbell and Bozorgnia (2019) show that correlation between CAV and $\text{PSA}(T)$ is

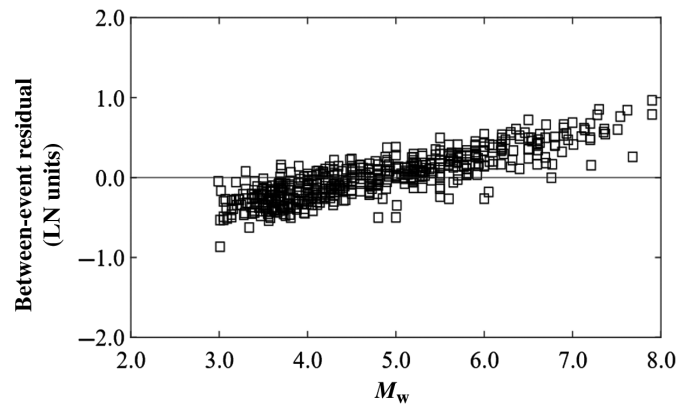


Figure 5. Residuals of equation (11) against M_w .

the highest for $T = 0$ (PGA). We can use empirical data and the point-source stochastic model to understand the expected scaling of MAA versus the PGA. Figure 4 shows the MAA/PGA ratio as a function of magnitude for different distances based on the point-source stochastic model, along with empirical-based trends, using recorded ground motions, in which case MAA/PGA was approximated as $\text{CAV}/(\text{PGA} \times D_{5-95})$. We observe that the empirical-based MAA/PGA ratio varies between 0.20 and 0.30, a range that includes the value of 0.22 from the point-source stochastic model. In addition, we observe that the MAA/PGA has some magnitude dependence, especially for distances larger than 5 km. Based on the discussion presented above, PGA should be a parameter to include in a CAV model, and Figure 4 suggests that M_w should be also included. In addition, as it is evident from equation (10), duration (or a proxy to duration) must be included.

We performed an additional regression using the functional form in equation (11), in which PGA, and D_{5-95} were included. The regression coefficients are $a_0 = -1.11$, $a_1 = 0.89$, and $a_2 = 0.56$, which show a strong dependence between CAV, PGA, and D_{5-95} , as expected. The residuals of this regression against M_w are presented in Figure 5, showing trends, confirming that M_w should be included in a conditional model for CAV. M_w is capturing average acceleration and duration effects not captured by the scaled $\ln(\text{PGA})$ and the $\ln(D_{5-95})$:

$$\ln(\text{CAV}) = a_0 + a_1 \ln(\text{PGA}) + a_2 \ln(D_{5-95}). \quad (11)$$

To understand the importance of other parameters on estimating CAV, we performed an analysis of feature selection (including M_w , R_{rup} , V_{S30} , and $\text{PSA}(T)$, with 22 values for T from 0.1 to 5 s) using the following machine-learning algorithms: (1) forward stepwise selection (FSS) and (2) least absolute shrinkage and selection operator (LASSO; Tibshirani, 1996). Figure 6 shows the dependency of model accuracy on the number of features selected by FSS. It can be observed

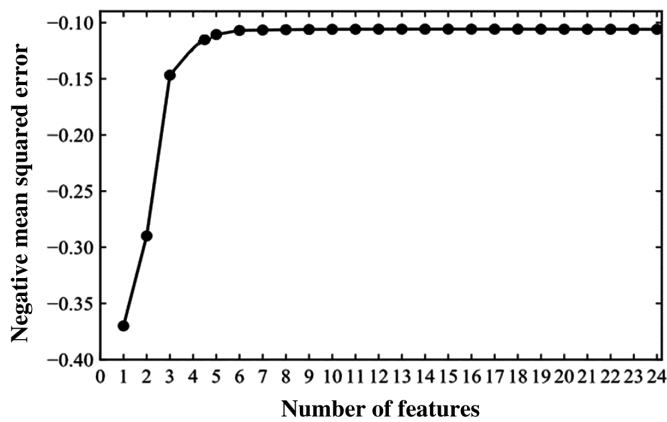


Figure 6. Results of the forward stepwise selection. Beyond five features (M_w , R_{rup} , V_{S30} , PGA, and $PSA(T = 4.0 \text{ s})$), there is negligible gain on performance. PSA , pseudospectral acceleration.

that the cross-validated negative mean-squared error barely increases, when the number of features exceeds five (it stabilizes at 0.101). This result implies that a conditional CAV model, including around five features, will achieve significant performance, and there is a negligible gain on performance by including more features. The five features selected by the FSS method were M_w , R_{rup} , V_{S30} , PGA, and $PSA(T = 4.0 \text{ s})$. LASSO selected many more features (M_w , R_{rup} , V_{S30} , PGA, $PSA(T = 0.2 \text{ s})$, $PSA(T = 1.0 \text{ s})$, $PSA(T = 1.4 \text{ s})$, $PSA(T = 1.6 \text{ s})$, $PSA(T = 1.8 \text{ s})$, $PSA(T = 3.2 \text{ s})$, $PSA(T = 3.4 \text{ s})$, $PSA(T = 3.6 \text{ s})$, $PSA(T = 3.8 \text{ s})$, $PSA(T = 4.0 \text{ s})$) but with only a minor improved performance compared to FSS (a cross-validation error of 0.100 compared with the 0.101 on FSS); hence, confirming the features selected by FSS are optimal. In addition, when $PSA(T = 4.0 \text{ s})$ is removed, the cross-validation error increases only slightly to 0.106. Physically, R_{rup} is a well-known proxy to duration, and V_{S30} is capturing average acceleration and duration effects that depend on site conditions.

As a summary, we found that PGA, M_w , R_{rup} , V_{S30} are the optimal parameters for a CAV CGMM (as checked later, exclusion of $PSA(T = 4.0 \text{ s})$ does not introduce biases in the model). Recall that CAV represents the product of an average acceleration times duration. PGA is a proxy for the average acceleration, V_{S30} relates site conditions effects on the average acceleration and duration, M_w provides additional information on the average acceleration and duration, and, finally, R_{rup} is a proxy for duration.

CONDITIONAL CAV MODELS

Based on the discussions in the previous section, the following functional form is considered for the CAV CGMM, the estimate of CAV is conditioned on the PGA, M_w , R_{rup} , and V_{S30} , the last term is included to accommodate potential biases due to HW effects, as recommended by Abrahamson *et al.* (2016) who considered this terms on CGMMs for I_A :

$$\ln \left[\text{CAV} \left(\frac{\text{m}}{\text{s}} \right) \right] = c_1 + c_2 \ln \text{PGA} + c_3 M_w + c_4 \ln V_{S30} + c_5 \ln R_{rup} + c_6 F_{HW} T_1(\text{dip}) T_2(M_w) T_5(R_{JB}), \quad (12)$$

in which PGA is in g, V_{S30} is in m/s, and R_{rup} is in km. F_{HW} is a flag that is 1 for sites located on the HW side of the rupture and 0 for sites located on the FW side of the rupture. The other terms follow the functional forms of the dip, magnitude, and distance tapers in Abrahamson *et al.* (2014), and are given by equation (13a–c):

$$T_1(\text{dip}) = \begin{cases} (90 - \text{dip})/45 & \text{for dip} > 30 \\ 60/45 & \text{for dip} \leq 30 \end{cases}, \quad (13a)$$

$$T_2(M) = \begin{cases} 1 & \text{for } M \geq 6.5 \\ 1 + 0.2(M - 6.5) - 0.8(M - 6.5)^2 & \text{for } 5.5 < M < 6.5, \\ 0 & \text{for } M \leq 5.5 \end{cases} \quad (13b)$$

$$T_5(R_{JB}) = \begin{cases} 1 & \text{for } R_{JB} = 0 \\ 1 - \frac{R_{JB}}{15} & \text{for } R_{JB} < 15. \\ 0 & \text{for } R_{JB} \geq 15 \end{cases} \quad (13c)$$

We calculate the coefficients in equation (12) using random-effects regressions, which are often employed in developing GMMs, addressing the correlation in the data through the event terms (e.g., Abrahamson and Youngs, 1992). The CGMM developed in this study considers random effects across different earthquakes. The resulting coefficients are listed in Table 1. The total standard deviation for the developed CGMM is 0.35 in natural log units, which, as expected, is much lower than the traditional GMMs developed for CAV. For example, the recent non-CGMM for CAV by Campbell and Bozorgnia (2019) has a standard deviation ranging from 0.45 to 0.58.

The residuals of the CGMM are shown in Figure 7, as a function of the predictor parameters used in equation (12) and also other parameters of importance. The distribution of between-residuals with respect to magnitude (M_w) and the depth to the top of rupture (Z_{TOR}) are shown in Figure 7a,b. Within-event residuals are plotted against PGA, $PSA(T = 4.0 \text{ s})$, V_{S30} , and R_{rup} in Figure 7c–f. No systematic trends are found in the plots for PGA, $PSA(T = 4.0 \text{ s})$, V_{S30} , M_w , R_{rup} , indicating that the model is not biased with respect to these parameters. In addition, we did not observe trends in terms of other $PSA(T)$, not shown in Figure 7.

TABLE 1
Results from the Random-Effects Regression

Coefficients						Error Terms	
c1	c2	c3	c4	c5	c6	τ	ϕ
1.79	0.67	0.57	−0.47	−0.0026	0.17	0.17	0.26

c1, c2, c3, c4, and c5 are coefficients in equation (2). ϕ is the within-event error term, and τ is the between-event error term.

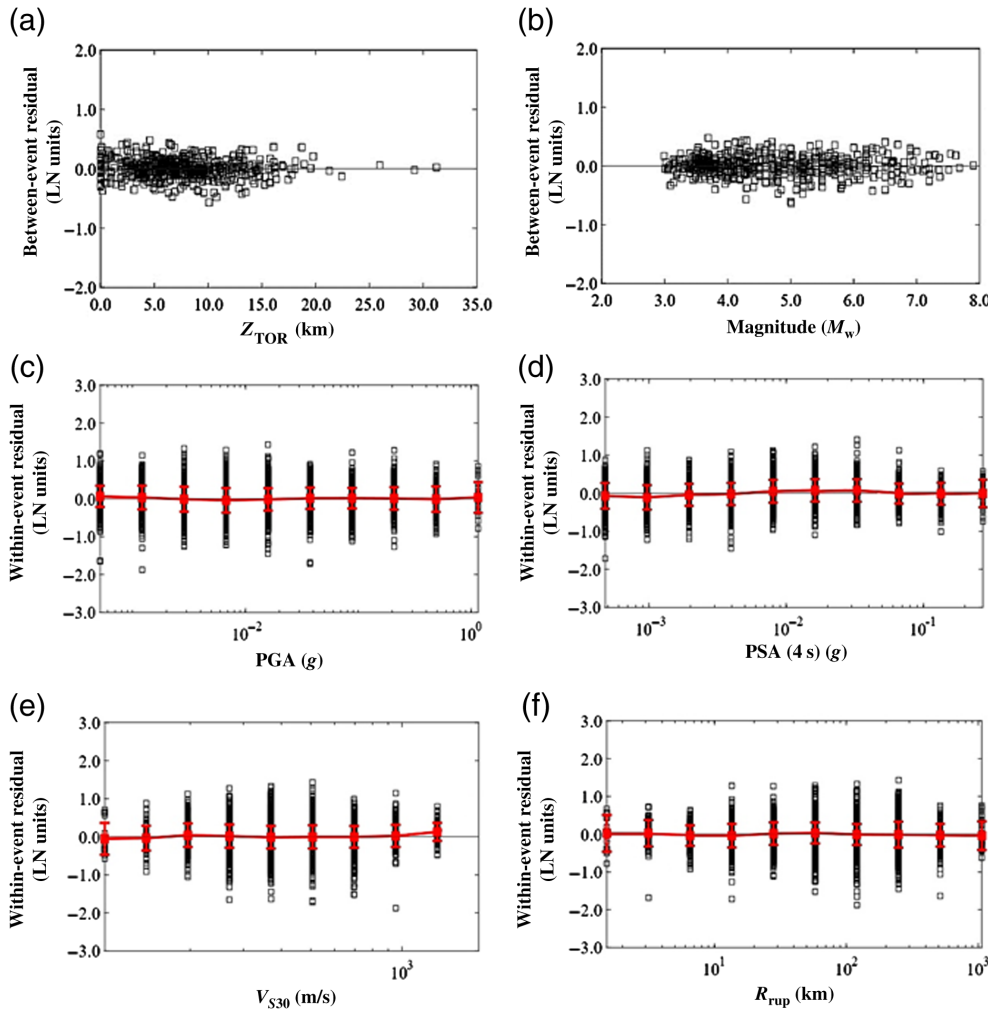


Figure 7. Residuals of the conditional model for CAV. Between residuals for (a) Z_{TOR} and (b) M_w . Within residuals for (c) PGA, (d) $PSA(T = 4.0 \text{ s})$, (e) V_{S30} , and (f) R_{rup} . The color version of this figure is available only in the electronic edition.

CONVERSION OF THE CONDITIONAL MODELS TO SCENARIO-BASED MODELS

By combining the CGMM in the previous section with traditional non-CGMMs for PGA, a suite of scenario-based models can be established. Five scenario-based models for CAV are developed using the GMMs for PGA from the NGA-West2 project. The models considered are: ASK14 (Abrahamson *et al.*, 2014), BSSA2014 (Boore *et al.*, 2014), CB2014 (Campbell and Bozorgnia, 2014), CY2014 (Chiou and Youngs, 2014), and I2014 (Idriss, 2014).

The median of CAV for a given earthquake scenario can be estimated using the following equation:

$$\ln \left[\text{CAV}_{\text{med}} \left(\frac{\text{m}}{\text{s}} \right) \right] = c1 + c2 \ln \text{PGA}_{\text{med}} + c3 M_w + c4 \ln V_{S30} + c5 \ln R_{rup} + c6 F_{\text{HW}} T_1(\text{dip}) T_2(M_w) T_5(R_{JB}), \quad (14)$$

in which CAV_{med} is the median estimate in m/s, PGA_{med} is the median estimate given by the traditional GMM being considered in g .

For instance, the PGA_{med} can be estimated through equation (15), when the ASK14 model is used:

$$\begin{aligned} \ln[\text{PGA}(g)] = & f_1(M, R_{rup}) \\ & + F_{\text{RV}} f_7(M) + F_{\text{N}} f_8(M) \\ & + F_{\text{AS}} f_{11}(\text{CR}_{JB}) \\ & + f_5(\widehat{\text{PGA}}_{1180}, V_{S30}) \\ & + F_{\text{HW}} f_4(R_{JB} + R_{rup}) \\ & + R_x, R_{y0}, W, \text{dip}, Z_{TOR}, M \\ & + f_6(Z_{TOR}) + f_{10}(Z_1, V_{S30}) \\ & + \text{Regional}(V_{S30}, R_{rup}), \quad (15) \end{aligned}$$

in which M is the magnitude, R_{rup} is the rupture distance, dip is the fault dip in degrees, W is the fault width, $\widehat{\text{PGA}}_{1180}$ stands for the median peak spectral acceleration for $V_{S30} = 1180 \text{ m/s}$, R_{JB} is the Joyner–Boore distance, CR_{JB} is the centroid R_{JB} , R_x is the horizontal distance from top of the rupture, R_{y0} is the horizontal distance off the end of the rupture measured parallel to strike, Z_{TOR} is the depth to

the top of rupture, $f_1, f_4, f_5, f_6, f_7, f_8, f_{10}, f_{11}$, and “Regional” are functions to control the magnitude scaling, nonlinear site effects, depth scaling, distance scaling, and regionalization proposed by Abrahamson *et al.* (2014). F_{RV} and F_{N} are flags for reverse- and normal-faulting earthquakes, respectively, F_{HW} is the flag for HW sites, and F_{AS} is the flag for aftershocks.

Given the standard deviation for the PGA GMMs, the standard deviation for the scenario-based models can be estimated, using propagation of errors (Bevington and Robinson, 1969), using the following equation:

$$\sigma_{\ln \text{CAV}}^2 = \sigma_{\ln \text{CAV} | \ln(\text{PGA})}^2 + \sigma_{\ln \text{PGA}}^2 [\partial \ln \text{CAV} / \partial \ln \text{PGA}]^2. \quad (16)$$

The partial derivative can be calculated using equation (12) (i.e., the conditional model), which leads to $\partial \ln \text{CAV} / \partial \ln \text{PGA} = c2$. Hence, the total standard deviation of the

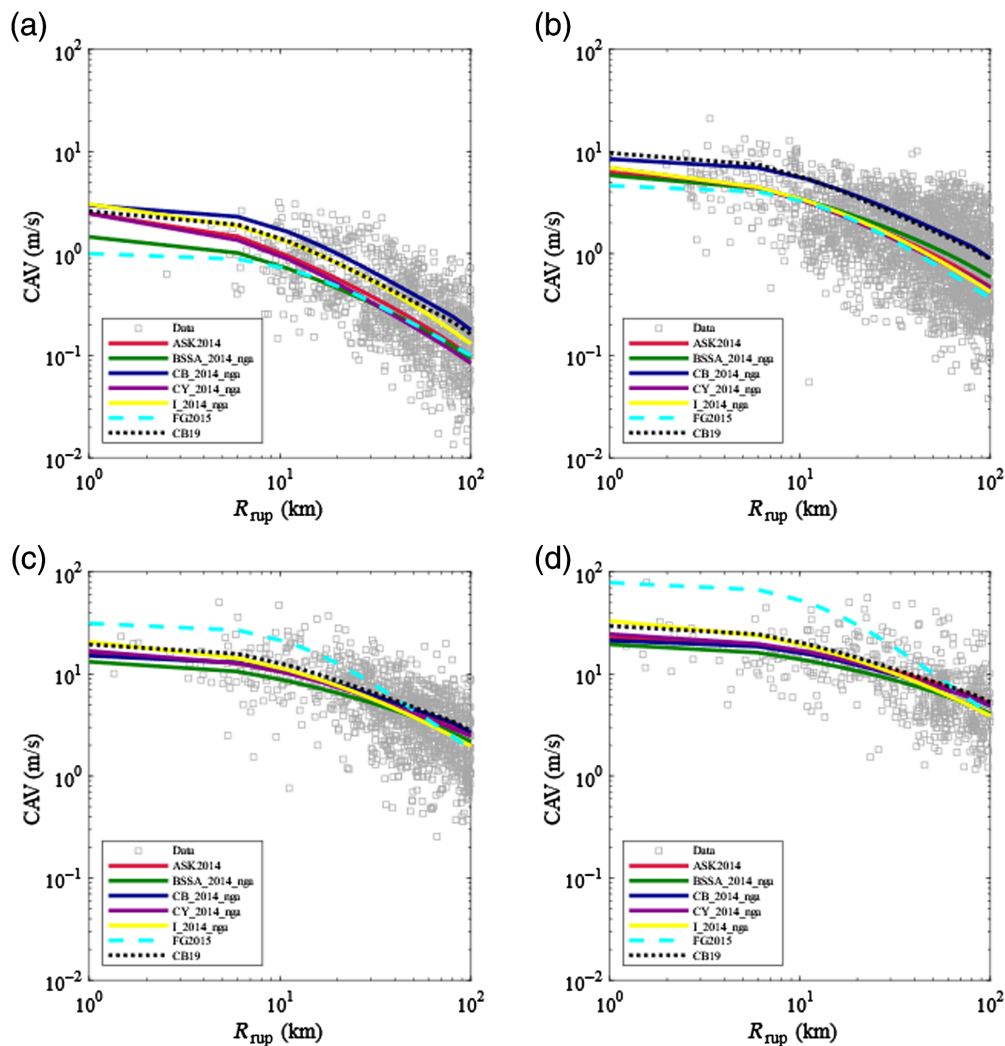


Figure 8. Comparisons of distance scaling of median CAV estimate for scenario-based CAV models developed in this study and for nonconditional CAV ground-motion models (GMMs). (a) $M_w = 4.75$, $V_{S30} = 424$ m/s, (b) $M_w = 5.75$, $V_{S30} = 424$ m/s, (c) $M_w = 7$, $V_{S30} = 424$ m/s, and (d) $M_w = 7.6$, $V_{S30} = 424$ m/s. Spectral acceleration models: ASK14 (Abrahamson *et al.*, 2014), BSSA2014 (Boore *et al.*, 2014), CB2014 (Campbell and Bozorgnia, 2014), CY2014 (Chiou and Youngs, 2014), and I2014 (Idriss, 2014). Nonconditional CAV GMMs: FG2015 (Foulser-Piggott and Goda, 2015) and CB19 (Campbell and Bozorgnia, 2019). The color version of this figure is available only in the electronic edition.

scenario-based model can be calculated according to the following equation:

$$\sigma_{\ln \text{CAV}} = \sqrt{0.097 + 0.4541 \times \sigma_{\ln \text{PGA}}^2}, \quad (17)$$

in which $\sigma_{\ln \text{PGA}}$ is the standard deviation given by the GMMs used for estimating the median PGA.

MODEL TRENDS AND COMPARISONS WITH PREVIOUS CAV MODELS

The five scenario-based models for CAV are compared with existing non-CGMMs for CAV to assess model trends. The considered non-CGMMs correspond to: FG2015 (Foulser-Piggott and Goda, 2015), Bea2017 (Bullock *et al.*, 2017),

and CB19 (Campbell and Bozorgnia, 2019). The Bea2017 model has been formulated only for rock conditions; hence, it is not used in conditions different from rock. The scenario-based models are developed using the coefficients listed in Table 1, the mean estimates are calculated according to equation (14) and the standard deviation according to equations (16) and (17).

Figure 8 shows the distance scaling comparisons for the developed scenario-based CAV models, considering different earthquake scenarios (M_w values of 4.8, 5.8, 7.0, 7.6), a $V_{S30} = 424$ m/s, and a hypocenter depth of 15 km. Whereas Figure 9 shows trends for similar conditions but considering a $V_{S30} = 760$ m/s to include the Bea2017 model in the comparisons.

The distance scaling of all the developed scenario-based CAV models in Figures 8 and 9 is generally consistent. The CAV estimates in the scenario-based model derived using the CB2014 model are slightly higher than the estimates from other scenario-based models for low-to-moderate magnitudes (e.g., $M_w = 4.8, 5.8$) and large V_{S30} (i.e., $V_{S30} = 760$ m/s),

whereas, at larger magnitudes (e.g., $M_w = 7.0, 7.6$), the estimates from all the scenario-based models are generally consistent. In terms of the nonconditional CAV GMM, the scaling of the CB19 model is similar to the scaling of the scenario-based models, whereas the scaling in the FG2015 model is stronger at short distances (i.e., $R_{\text{rup}} \leq 30$ km) and large magnitudes (i.e., $M_w > 6.0$), and the scaling of the Bea2017 model for $R_{\text{rup}} \leq 10$ km is stronger compared with the scaling in the scenario-based models. This occurs for the two V_{S30} values being considered in Figures 8 and 9. The CAV estimates from the CB19 model are generally consistent with those from the scenario-based models, and, in particular, they are quite close to the estimates produced by the scenario-based model based on the CB_2014_rga GMM. The estimates from the FG2015

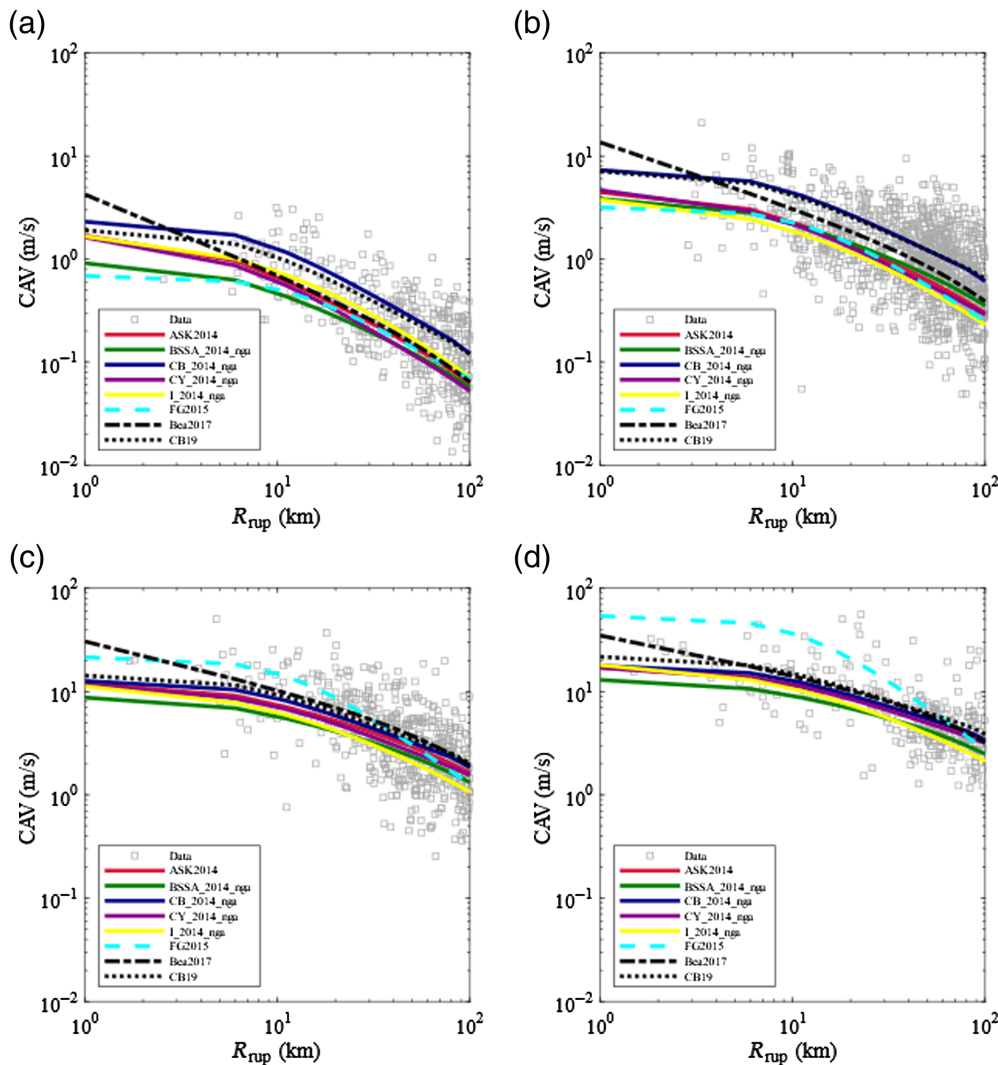


Figure 9. Comparisons of distance scaling of median CAV estimate for scenario-based CAV models developed in this study and for nonconditional CAV GMMs. (a) $M_w = 4.75$, $V_{S30} = 760$ m/s, (b) $M_w = 5.75$, $V_{S30} = 760$ m/s, (c) $M_w = 7$, $V_{S30} = 760$ m/s, and (d) $M_w = 7.6$, $V_{S30} = 760$ m/s. Spectral acceleration models: ASK14 (Abrahamson *et al.*, 2014), BSSA2014 (Boore *et al.*, 2014), CB2014 (Campbell and Bozorgnia, 2014), CY2014 (Chiou and Youngs, 2014), and I2014 (Idriss, 2014). Nonconditional CAV GMMs: FG2015 (Foulser-Piggott and Goda, 2015), Bea2017 (Bullock *et al.*, 2017), and CB19 (Campbell and Bozorgnia, 2019). The color version of this figure is available only in the electronic edition.

model are generally slightly lower than those in other models (including the CB19 and Bea2017 models) for low-to-moderate magnitudes (i.e., at $M_w = 4.8, 5.8$) and higher for large magnitudes (i.e., at $M_w = 7.0, 7.6$). This occurs for the two V_{S30} values being considered in Figures 8 and 9. The differences at large magnitudes and short distances may be associated with the treatment of the short distance saturation in the FG2015 model. The estimates from the Bea2017 model in Figure 9 are higher compared with the estimates from other models, when $R_{rup} \leq 10$ km and $M_w = 4.8, 5.8$. For large magnitudes ($M_w = 7.0, 7.6$), the estimates from the Bea2017 model are still larger than those from other models, with exception of the FG2015 model. For $R_{rup} > 10$ km, the estimates of the

Bea2017 model are generally consistent with those from the implemented scenario-based models.

The variation of the standard deviation versus rupture distance, for the scenario-based models and non-CGMMs, are compared in Figure 10. The comparisons consider different earthquake scenarios (M_w values of 4.8, 5.8, 7.0, 7.6), a $V_{S30} = 424$ m/s, and a hypocenter depth of 15 km. The FG2015 GMM has the largest standard deviation (0.64 in ln units) at all earthquake scenarios. The scenario-based models have similar standard deviations, varying from 0.42 to 0.51 in ln units, which are also similar to the standard deviation in CB19 model. The scenario-based model based on the CB_2014 GMM has the lowest standard deviation in most earthquake scenarios, whereas the scenario-based model based on the I_2014 GMM has the largest standard deviation.

The magnitude scaling for the developed scenario-based models and non-CGMMs are compared in Figures 11 and 12, considering different earthquake scenarios R_{rup} values of 5 and 10 km, site conditions ($V_{S30} = 424$ m/s, 760 m/s), and a hypocenter depth of 15 km. The magnitude scaling

across the different scenario-based models is generally consistent, especially for large magnitudes (i.e., $M_w > 6.0$). In terms of the traditional non-CGMMs, the magnitude scaling in the CB19 and Bea2017 is consistent with the scaling in the scenario-based models, whereas the FG2015 model shows a much stronger scaling at large magnitudes (i.e., $M_w > 6.0$). The estimates from all scenario-based models and non-CGMMs are generally consistent with the exception of the FG2015 model, which provides larger estimates at large magnitudes. The consistency or discrepancy across the different scenario-based models and nonconditional models for large magnitudes and short distances depends on the treatment of magnitude saturation.

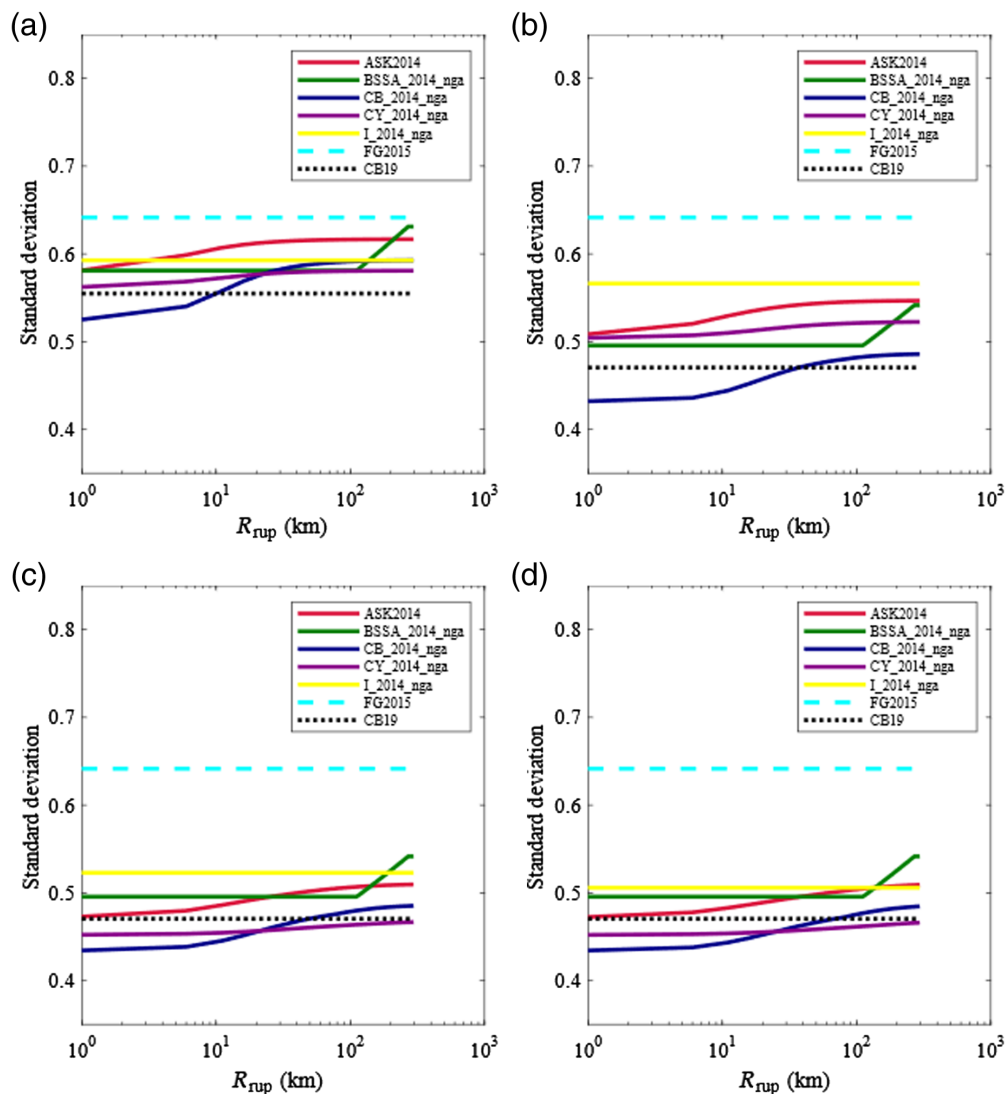


Figure 10. Comparisons of standard deviations against rupture distance of the five scenario-based models developed in previous section and nonconditional models for CAV. (a) $M_w = 4.75$, $V_{S30} = 424$ m/s, (b) $M_w = 5.75$, $V_{S30} = 424$ m/s, (c) $M_w = 7$, $V_{S30} = 424$ m/s, and (d) $M_w = 7.6$, $V_{S30} = 424$ m/s. Spectral acceleration models: ASK14 (Abrahamson *et al.*, 2014), BSSA2014 (Boore *et al.*, 2014), CB2014 (Campbell and Bozorgnia, 2014), CY2014 (Chiu and Youngs, 2014), and I2014 (Idriss, 2014). Nonconditional CAV GMMs: FG2015 (Foulser-Piggott and Goda, 2015) and CB19 (Campbell and Bozorgnia, 2019). The color version of this figure is available only in the electronic edition.

Figure 13 shows the variation of the standard deviation in terms of M_w , across different scenario-based and non-CGMMs. The FG2015 model shows the largest standard deviation (0.65 in ln units), whereas the standard deviations from the CB19 and the scenario-based models show comparable standard deviations (in the range of 0.5–0.6).

Interestingly, there is a remarkable consistency (in terms of trends and estimates) between the CAV scenario-based model implemented using the CB_2014_nga spectral acceleration GMM and the nonconditional CAV CB19 GMM (e.g., Figs. 9 and 12). In this case, both the scenario-based model and the CB19 GMM have been developed with the NGA-West2

database, and the underlain CB2014-nga model for the scenario-based implementation has similar constraints as the CB19 GMM (the two models were actually developed by the same authors).

Hence, scenario-based models can be as robust as traditional non-CGMM, if the underlain spectral acceleration GMM used to implement a scenario-based model is also robust. As previously discussed, this is an important advantage to develop GMMs for secondary IMs, because this approach takes advantage of the efforts invested in constraining the extrapolation of spectral acceleration GMMs. As we discussed earlier (i.e., Fig. 1), I_A is another IM that combines duration and amplitude information; hence, we derived a conditional model for I_A using the functional form in equation (12). The details of this model and its comparison against the trends observed for CAV are included in the supplemental material to this article. Of note, Abrahamson *et al.* (2016) also developed a CGMM for I_A , but with a slightly different functional form compared with the one used in this study.

PROBABILISTIC CAV HAZARD AT LOW-PROBABILITY LEVELS

Recent studies have shown that the tails from the residual's distribution of GMMs for spectral accelerations, which are associated with low-probability levels, tend to depart from the normal distribution (e.g., GeoPentech, 2015; Bullock, 2019; Kotha *et al.*, 2020). For example, GeoPentech (2015) found that the residuals from the NGA-West2 GMMs showed a departure from lognormal above two standard deviations, with more recordings at high-epsilon levels than would be predicted by a lognormal distribution. When the tails of the residual's distribution from a GMM produce more high-epsilon values than expected from a normal distribution, the behavior is known as fat tails (Abrahamson *et al.*, 2016). The normality

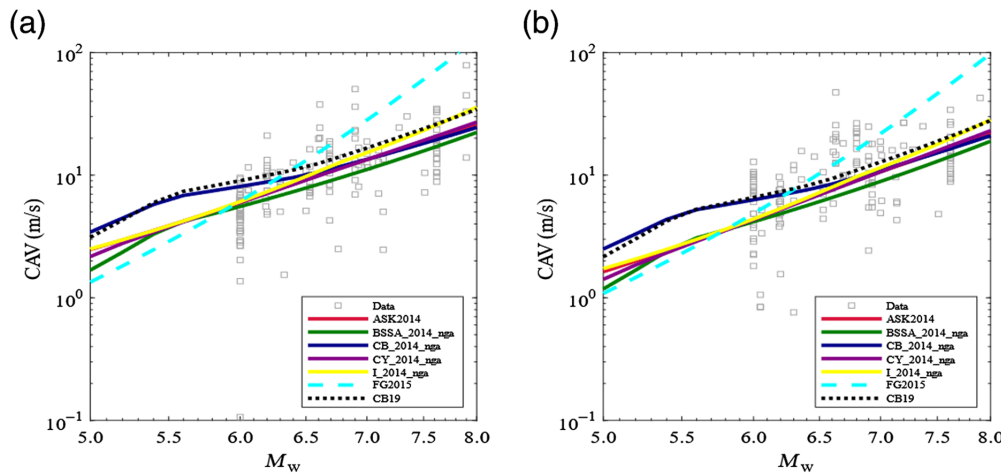


Figure 11. Comparisons of magnitude scaling for estimates of median CAV by scenario-based models and nonconditional models. (a) Rupture distance 5 km, $V_{S30} = 424$ m/s, and (b) rupture distance 10 km, $V_{S30} = 424$ m/s. Spectral acceleration models: ASK14 (Abrahamson *et al.*, 2014), BSSA2014 (Boore *et al.*, 2014), CB2014 (Campbell and Bozorgnia, 2014), CY2014 (Chiu and Youngs, 2014), and I2014 (Idriss, 2014). Nonconditional CAV GMMs: FG2015 (Foulser-Piggott and Goda, 2015) and CB19 (Campbell and Bozorgnia, 2019). The color version of this figure is available only in the electronic edition.

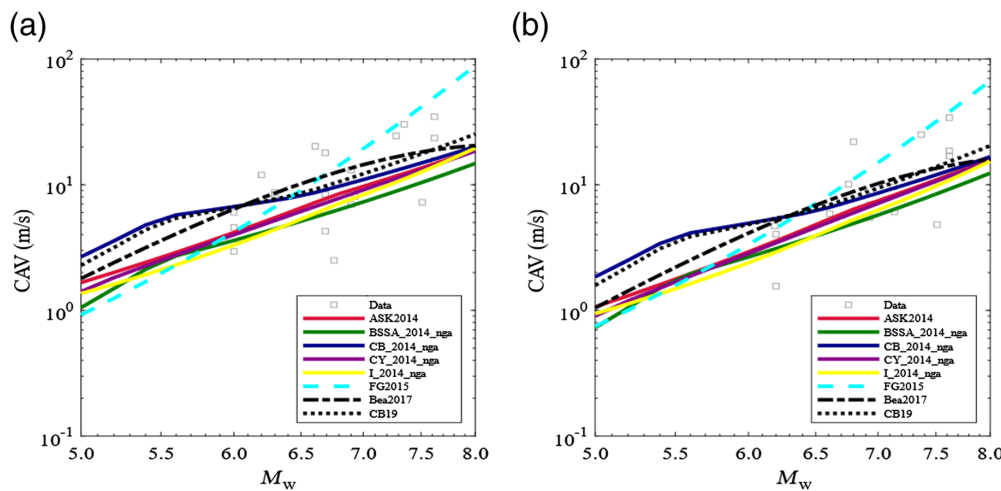


Figure 12. Comparisons of magnitude scaling for estimates of median CAV by scenario-based models and nonconditional models. (a) Rupture distance 5 km, $V_{S30} = 760$ m/s, and (b) rupture distance 10 km, $V_{S30} = 760$ m/s. Spectral acceleration models: ASK14 (Abrahamson *et al.*, 2014), BSSA2014 (Boore *et al.*, 2014), CB2014 (Campbell and Bozorgnia, 2014), CY2014 (Chiu and Youngs, 2014), and I2014 (Idriss, 2014). Nonconditional CAV GMMs: FG2015 (Foulser-Piggott and Goda, 2015), Bea2017 (Bullock *et al.*, 2017), CB19 (Campbell and Bozorgnia, 2019), and Bea2017 (Bullock *et al.*, 2017). The color version of this figure is available only in the electronic edition.

assumption is checked using normal probability plots in Figure 14. The distribution of between-event residuals is fairly normal, and the distribution of within-event residuals is consistent with a normal distribution between the 1% and 99% levels (e.g., about ± 2.3 standard deviations). GeoPentech (2015) modeled the fat tails for the within-event residuals using

a mixture model given by the weighted average of two normal distributions with the same mean of zero but different standard deviations. With this mixture model, the conditional probability of exceeding the ground motion is given in the following equation:

$$P(\text{CAV} > z | \text{PGA}, R_{\text{rup}}, M, V_{S30}) = \Pi_1 \left(1 - \Phi \left(\frac{\ln z - \ln \text{CAV}}{\sqrt{\phi_1^2 + \tau^2}} \right) \right) + \Pi_2 \left(1 - \Phi \left(\frac{\ln z - \ln \text{CAV}}{\sqrt{\phi_2^2 + \tau^2}} \right) \right). \quad (18)$$

In the previous equation, both distributions have the mean of 0 with $\Pi_1 = 0.42$, $\phi_1 = 0.270$ and $\Pi_2 = 0.576$, $\phi_2 = 0.333$ respectively. For both distributions, $\tau = 0.18$. The mixture model is able to effectively capture the fat tails, as shown in Figure 15.

USE OF THE CGMM AND SCENARIO-BASED MODELS IN ENGINEERING PRACTICE

This section presents potential applications of the models implemented in this study on engineering practice. We discuss in the following the application for PSHA. Applications to calculate coefficients of correlation and selection of ground motions are discussed in the supplemental material.

PSHA

The implemented scenario-based CAV GMMs can be

employed to perform PSHAs for the evaluation of CAV hazard curves. In general IM hazard curves are employed as inputs into performance based procedures (e.g., Macedo *et al.*, 2018). The annual rate of exceedance for a given CAV_z threshold can be estimated as (in cases where fat tails are ignored):

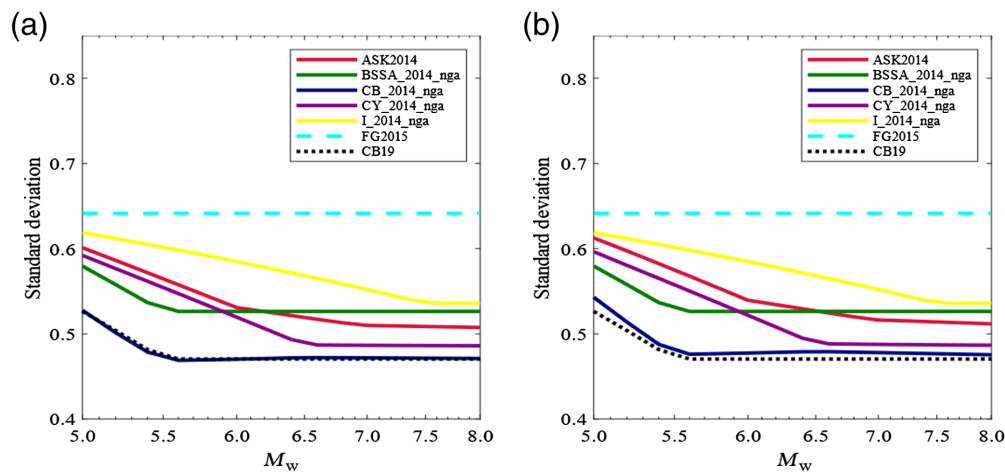


Figure 13. Comparisons of standard deviations against magnitude of the five scenario-based models developed in previous section and nonconditional models for CAV. (a) Rupture distance 5 km, $V_{S30} = 424$ m/s, and (b) rupture distance 10 km, $V_{S30} = 424$ m/s. Spectral acceleration models: ASK14 (Abrahamson *et al.*, 2014), BSSA2014 (Boore *et al.*, 2014), CB2014 (Campbell and Bozorgnia, 2014), CY2014 (Chiou and Youngs, 2014), and I2014 (Idriss, 2014). Nonconditional CAV GMMs: FG2015 (Foulser-Piggott and Goda, 2015) and CB19 (Campbell and Bozorgnia, 2019). The color version of this figure is available only in the electronic edition.

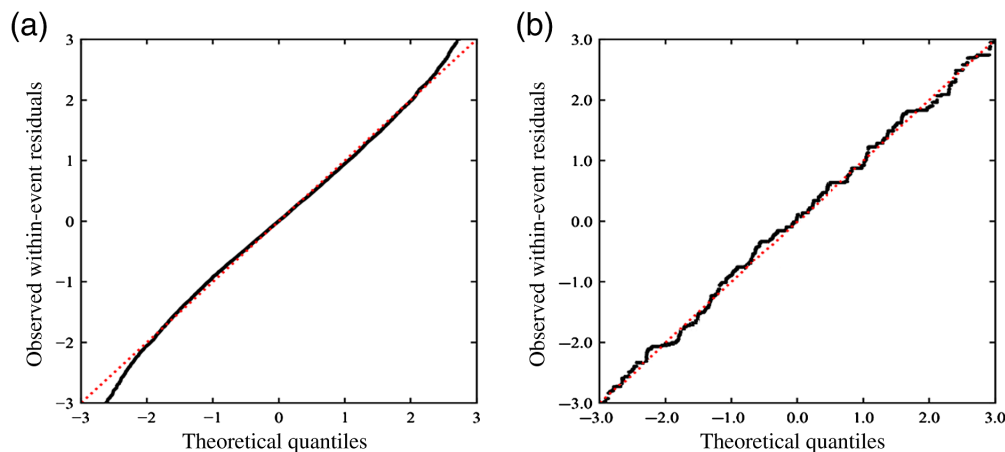


Figure 14. Normal probability plot for (a) within-event residuals and (b) between-event residuals. The color version of this figure is available only in the electronic edition.

$$P(\text{CAV} > \text{CAV}_z | \text{PGA}, M, V_{S30}) = 1 - \Phi \left[\frac{\text{CAV}_z - \ln(\text{CAV}_{\text{med}})}{\sigma_{\ln \text{CAV}}} \right], \quad (19)$$

in which CAV_{med} and $\sigma_{\ln \text{CAV}}$ are the median and standard deviation estimates in the scenario-based models, which can be evaluated using equations (14)–(17). In the following, we illustrate the evaluation of a CAV hazard curve by considering a simplified example with one seismic source.

The seismic source is assumed to be a strike-slip fault situated in a shallow crustal tectonic setting (e.g., California). The fault has the following characteristics: length of 25 km,

crustal thickness of 12 km, a truncated exponential model with a b -value (slope) of 0.9, a minimum magnitude of 5.0, a maximum magnitude of 6.5, and an activity rate of 0.14. The evaluation point is assumed to be located at a distance of 13 km from the seismic fault. Figure 16a shows a schematic representation of the fault and the evaluation point. The CAV_z threshold values are from 0.001 to 50 m/s. The PSHA evaluation has been performed with the software SeismicHazard (Candia *et al.*, 2018, 2019), in which the CAV models developed in this study have been implemented. Figure 16b shows the CAV hazard curve, using the scenario-based model based on the ASK (Abrahamson *et al.*, 2014) GMM).

CONCLUSIONS

This study derives a new conditional CAV model and new scenario-based CAV models for shallow crustal earthquake zones, using the PEER NGA-West2 database for shallow crustal ground motions. We showed that CAV is a high-frequency parameter that is only weakly affected by the low-frequency content that controls the velocity range. Hence, CAV has little to do with velocity, despite its name. Because

CAV represents the product of an average acceleration times duration, we conditioned the CAV CGMM on PGA, which is a proxy for the average acceleration, V_{S30} , which relates site conditions effects on the average acceleration and duration, M_w , which provides additional information on the average acceleration and duration, and, finally, R_{rup} , which is a proxy for duration.

The CGMM provides robust CAV estimates that are consistent with a design spectral acceleration response spectrum, which may not be the case if nonconditional traditional GMM are employed for CAV. Hence, the primary use of the derived CGMM for CAV in engineering practice should

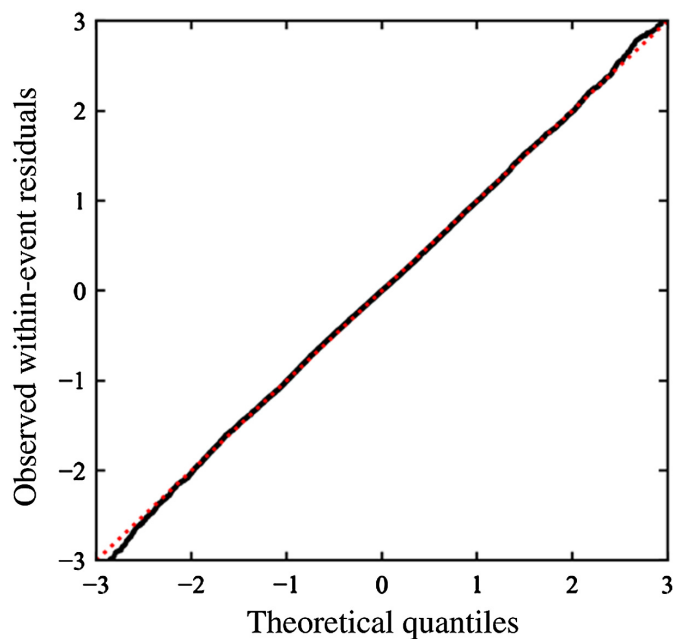


Figure 15. Normal probability plot using the mixture model for the conditional CAV model. The color version of this figure is available only in the electronic edition.

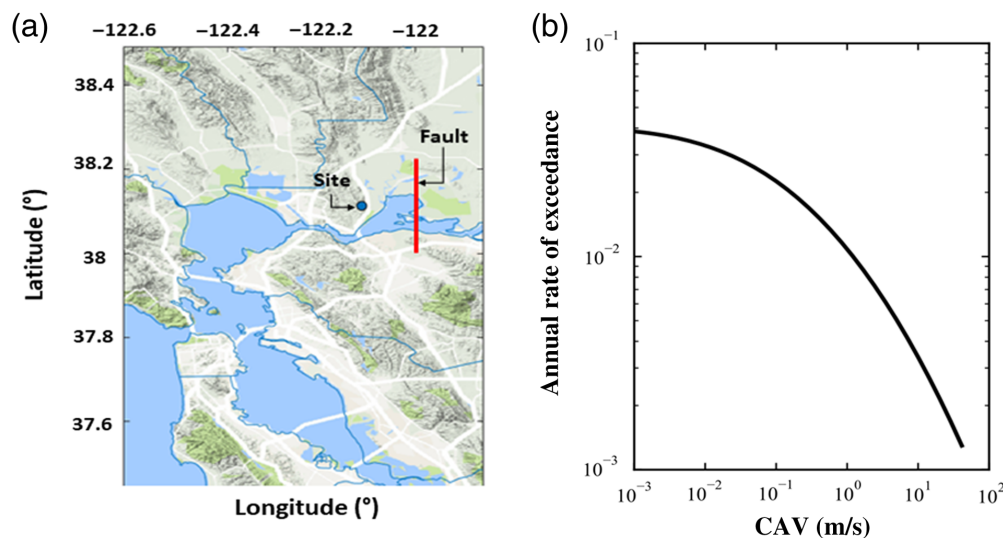


Figure 16. (a) Schematic representation of the strike-slip seismic fault and the evaluation site for the calculation of a CAV hazard curve and (b) estimated CAV hazard curve. The color version of this figure is available only in the electronic edition.

be to constrain the selection of ground motions to be consistent with a target design spectral acceleration response spectrum, when CAV is judged to be an important IM that needs to be included in the ground-motion selection.

Scenario-based models for CAV are derived by combining the conditional model with nonconditional models for spectral

accelerations (specifically for PGA). A scenario-based model can provide estimates for the median CAV and its standard deviation given the earthquake scenario (e.g., M_w , R_{rup}), site conditions, and other earthquake parameters. Hence, scenario-based models have all the features that nonconditional CAV models have, and they can also be readily used in performance-based earthquake engineering procedures that employ CAV. For instance, this study illustrated the use of scenario-based models to perform a PSHA assessment for CAV, the calculation of coefficients of correlation between CAV and spectral accelerations, and the selection of ground motions.

This study has developed five scenario-based models for CAV using the five GMMs for spectral accelerations from the NGA-West2 project. The scenario-based models scaling against magnitude and distance is generally consistent with available recorded ground-motion data. The scaling is also often consistent with the scaling in existing nontraditional CAV GMM, with the trends depending on the earthquake scenario and site conditions. When there are important discrepancies between the scenario-based CAV models and the considered nonconditional models, the scenario-based models are more consistent with the recorded ground-motion data and have reasonable trends. Importantly, scenario-based GMMs are highly congruous with traditional well constrained non-CGMMs if the underlying spectral

acceleration GMM used to implement a scenario-based model is properly constrained. An important advantage of the scenario-based CAV models developed in this study is that they inherit the extrapolation constraints imposed in the underlain spectral accelerations GMMs, which are often based on physics-based simulations. This provides confidence in the use of the implemented scenario-based models in ranges with few data.

Finally, we recommend the use of the CAV scenario-based models implemented in this study, as alternative models to the existing traditional nonconditional CAV models to account for epistemic uncertainties in performance-based earthquake

engineering evaluations that employ CAV as an intensity measure.

DATA AND RESOURCES

The Pacific Earthquake Engineering Research Center (PEER) Next Generation Attenuation-West2 (NGA-West2) ground-motion

database for shallow crustal earthquakes is used in this study. The software R (<http://www.R-project.org/>, last accessed October 2019) was used to perform the regressions. The comparison with an I_A continuous ground-motion model (CGMM) and additional applications in engineering practice are included as a supplemental material to this article.

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